

■ NurseFlow — Enterprise System Architecture & API Documentation

Comprehensive System Reference Manual | Smart Duty Rostering, Gemma 4 AI Clinical Patient Simulator, and Shift Break Lounge Engine

1. Executive Overview & Problem Vision

NurseFlow is an enterprise-grade full-stack web platform engineered to modernize hospital workforce scheduling, eliminate shift overlaps, reduce clinical nurse burnout, and provide continuous communication skill training. Built with React 18, Node.js, Prisma ORM, and Google Gemma 4 AI, NurseFlow transforms hospital operational efficiency.

Primary Platform Pillar Goals:

- **Duty Roster Optimization:** Elimination of text collisions and scheduling friction through clear weekly/monthly card layouts.
- **AI Clinical Training (Gemma 4):** Interactive patient roleplay simulation enabling nurses to practice communicating with high-stress patient and caregiver personas.
- **Shift Break Fatigue Reduction:** 100% non-medical relaxation games (AI Tick & Cross ■■) to revitalize mental focus during 10-minute shift breaks.
- **Real-Time Event Dispatch:** WebSocket-based push notification engine for instant ward emergency broadcasts and shift swap approvals.

2. Full-Stack System Architecture

NurseFlow follows a decoupled client-server architecture. The frontend application communicates with the Node.js Express backend via REST API endpoints and Socket.IO WebSockets for real-time notifications.

Layer	Technology / Framework	Architecture Description & Purpose
Frontend Client	React 18, Vite, TypeScript, Tailwind CSS	Single-Page Application (SPA) with glassmorphism design tokens & responsive components.
State & Routing	React Context API, React Router v6	Global AuthContext, SocketContext, and protected route guards.
Backend Server	Node.js, Express, TypeScript	REST API routing, JWT authentication, and error-handling middleware.
Database Layer	Prisma ORM, SQLite (dev.db)	Relational data modeling, schema migrations, and type-safe query generation.
AI Intelligence	Google Gemma 4 AI Engine	Roleplay persona generation, 8-score analysis engine, and Minimax AI for Tic-Tac-Toe.

Real-Time Messaging	Socket.IO (WebSockets)	Bi-directional real-time alert broadcasts and unread badge sync.
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3. Core Module 1: Smart Duty Roster Calendar Engine

The Roster Calendar module (NurseRoster.tsx) presents nurses and hospital administrators with a clear, color-coded weekly and monthly duty grid view.

Standardized Shift Hours & Styling Tokens:

- **Morning Shift:** 7:00 AM – 3:00 PM (Soft Emerald Tint bg-emerald-50/70 border-emerald-200)
- **Evening Shift:** 3:00 PM – 11:00 PM (Soft Amber Tint bg-amber-50/70 border-amber-200)
- **Night Shift:** 11:00 PM – 7:00 AM (Soft Light Purple Tint bg-[#EDE9FE]/50 border-purple-200)
- **Off Duty:** No Duty Assigned (Soft Slate Tint bg-slate-50 border-slate-200)
- **Full-Width Action Button:** Every day card contains a full-width View Details → button that prevents text collision regardless of screen resolution.

4. Core Module 2: Gemma 4 AI Patient Communication Simulator

The AI Patient Communication Simulator allows nurses to practice Healthcare Communication with simulated patient and caregiver personas. It roleplays realistic situations to develop empathy, active listening, and de-escalation skills.

Simulation Engine Architecture & Features:

- **Pre-Seeded Scenarios:** *Anxious Patient, Angry Family Member, Confused Elderly Patient, Non-Cooperative Patient, Difficult Conversation, Emergency Communication.*
- **Dynamic Character Emotion Tracking:** Evaluates the nurse's tone in real time. Compassionate words ('understand', 'listen', 'help') trigger positive emotion transitions: **Hostile → Frustrated → Worried → Calm → Reassured.**
- **8-Competency Evaluation System:** Calculates 0–100 scores across *Empathy, Active Listening, Clarity, Professionalism, Emotional Intelligence, De-escalation, Patient Engagement, and Confidence.*
- **Actionable Coaching:** Generates transcript quote highlights (Effective vs Growth Opportunities) and personalized summary recommendations.

5. Core Module 3: Gemma 4 AI Tick & Cross (Tic-Tac-Toe ■■■) Break Game

To help nurses disconnect from shift stress during 10-minute breaks, NurseFlow includes a 100% non-medical **Tick & Cross (Tic-Tac-Toe) Break Game** powered by Gemma 4 AI.

- **3 AI Difficulty Modes:** Easy (Casual Chill), Medium (Smart Rival), and Unbeatable (Minimax Decision Tree Algorithm).
- **Scoreboard & Wellness Points:** Tracks Nurse Wins, AI Wins, Draws, and awards Wellness Points.
- **10-Minute Break Countdown:** Built-in live timer ensures shift breaks stay structured and restorative.

6. Database Schemas & Data Dictionary (Prisma ORM)

```
model User {
  id          String    @id @default(uuid())
  name        String
  email       String    @unique
  passwordHash String
  role        String    @default("NURSE") // NURSE, ADMIN, HEAD_NURSE
  employeeId  String    @unique
  departmentId String?
}

model Roster {
  id          String    @id @default(uuid())
  nurseId     String
  shiftId     String
  departmentId String
  date        String    // YYYY-MM-DD
  status      String    @default("SCHEDULED")
  notes       String?
}

model CommunicationSession {
  id          String    @id @default(uuid())
  userId      String
  scenarioId  String
  characterRole String
  difficulty  String    @default("BEGINNER")
  status      String    @default("ACTIVE")
  overallScore Int?
}

model NurseGameScore {
  id          String    @id @default(uuid())
  userId      String
  gameType    String    @default("TIC_TAC_TOE")
  nurseWins   Int        @default(0)
  aiWins      Int        @default(0)
  draws       Int        @default(0)
  pointsEarned Int        @default(0)
}
```

7. Complete Backend REST API Reference

Method	Endpoint Route	Functionality Description	Access
POST	/api/auth/login	Authenticate user credentials & return JWT token	Public
POST	/api/auth/register	Register new nurse or administrator account	Public
GET	/api/users	Fetch list of all staff nurses & employees	Admin
GET	/api/rosters	Fetch duty rosters (filter by date/nurse)	Auth
POST	/api/rosters	Assign shift roster entry to nurse	Admin
POST	/api/leave-requests	Submit leave application (Sick, Annual)	Nurse
POST	/api/shift-swaps	Request shift exchange with target nurse	Nurse
POST	/api/notifications/send-alert	Broadcast ward emergency push notification	Admin
GET	/api/communication-simulator/scenarios	Fetch roleplay scenarios list	Auth
POST	/api/communication-simulator/sessions	Initialize new AI roleplay session	Auth
POST	/api/communication-simulator/sessions/:id/messages	Send nurse dialogue & get AI roleplay reply	Auth
POST	/api/communication-simulator/sessions/:id/end	End session & generate 8-competency evaluation	Auth
POST	/api/games/tic-tac-toe/move	Compute Gemma 4 AI move (Minimax)	Auth
POST	/api/games/tic-tac-toe/score	Save match outcome & update wellness points	Auth

8. Operational Setup & Deployment Guide

Local Setup Commands:

```
# 1. Install Server & Client Dependencies
cd server && npm install
cd ../client && npm install

# 2. Sync Prisma Database Schema & Run Seeding
cd ../server
npx prisma db push
npx ts-node prisma/seed.ts

# 3. Start Backend Server (Port 5000) & Client (Port 5173)
npm run dev # inside server
npm run dev # inside client
```

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